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The influence of an alongshelf current on the formation and offshore transport of dense water from a coastal polynya

David C. Chapman

Department of Physical Oceanography, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

Abstract. The influence of an ambient alongshelf current on the formation and offshore transport of dense water from an idealized coastal polynya is examined by using a primitive-equation numerical model. Brine rejection during ice formation within the polynya is modeled by a prescribed surface buoyancy flux over a limited region along a straight coastline. A uniform alongshelf current is imposed at one end of the model domain. The basic ocean response is unchanged by the ambient current; i.e., an unstable density front forms around the polynya perimeter and breaks up into small-scale (15–30 km) eddies that carry the dense water away from the polynya. The ambient current continuously carries water out of the polynya region, thereby both increasing the alongshelf density flux and reducing the typical density anomaly generated. The total volume of dense water formed is not appreciably altered. The influence of a submarine canyon located downstream of the polynya (in the direction of the imposed alongshelf current) is examined. Almost no dense water enters a canyon oriented normally or diagonally to the coast because the ambient flow tends to follow isobaths and carries virtually all of the dense water around and past the canyon. An appreciable fraction of densified water may flow down a canyon oriented parallel to the coast, the details depending on the canyon geometry and the strength of the ambient current. Thus both the density anomalies produced by coastal polynyas and the pathways of dense water transport are sensitive to ambient shelf currents, potentially providing a source of interannual variability in dense water formation and export from the shelf.

1. Introduction

Several recent numerical modeling studies have investigated the formation and offshore transport of dense water from an idealized coastal polynya [*Gawarkiewicz and Chapman*, 1995; *Chapman and Gawarkiewicz*, 1995, 1997; *Chapman*, 1999]. In all cases, dense water is transported offshore by means of small-scale (15–30 km) eddies that are produced by instabilities of the geostrophic current around the polynya perimeter. Typically, the offshore eddy flux of dense water eventually balances the formation of new dense water, and so the density anomaly within the polynya reaches a quasi-steady equilibrium. The time to reach equilibrium, the equilibrium density anomaly, and the offshore flux of dense water can be estimated by simple algebraic formulas that depend on the polynya geometry and ice production rate (i.e., surface buoyancy flux).

All of these studies have assumed that the ocean is initially at rest with no ambient currents. However, many continental shelves have mean alongshelf currents, typically flowing cyclonically (shallow water on the right in the Northern Hemisphere) with speeds of 0.05–0.1 m s^{-1}, which could clearly alter dense water formation. For example, ambient currents would reduce the time that a water parcel is exposed to surface buoyancy fluxes in the polynya, thus reducing its density anomaly. If the ambient current is fast enough, then the maximum time that a water parcel remains beneath the polynya may be less than the time to reach the quasisteady state; so the equilibrium density anomaly may never be achieved. A more subtle effect may be in altering the instabilities of the geostrophic rim current by changing the local relative vorticity, which could change the dominant eddy size and/or strength or perhaps even stabilize the front.

An ambient alongshelf current also raises the possibility that dense water could be transported along the shelf and into the proximity of submarine canyons where it might descend more rapidly and leave the shelf. If true, this would support the notion that canyons, which are found on many high-latitude shelves where dense water is formed, are preferred locations of offshelf transport of dense shelf water. An observational example of this

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effect has been reported by *Weingartner et al.* [1998]. Using an extensive set of observations from the Chukchi Shelf, they found that strong northeast winds diverted the Bering Strait inflow to the western shelf for part of the winter, thereby weakening the flow over the northeast part of the shelf. The winds also opened large polynyas where hypersaline shelf waters were created, some of which moved offshore across isobaths. However, when the winds relaxed, the alongshelf flow to the northeast swept much of the dense water down Barrow Canyon and into the deep Arctic basin.

On the other hand, model studies which include submarine canyons but assume no ambient currents suggest that canyons do not preferentially draw dense water offshore. That is, canyons are passive: if dense water finds its way into a canyon, then it is channeled along the canyon axis as a slow-moving gravity current [e.g., *Chapman and Gawarkiewicz*, 1995; *Jiang and Garwood*, 1995]. This implies that, without ambient currents, the canyon must be located near the polynya to impact the cross-shelf transport of dense water. The model response in the presence of both ambient currents and canyons has not yet been studied.

With these results and ideas in mind, the influence of an ambient alongshelf current on the formation and offshore transport of dense water from an idealized coastal polynya is examined here by using a primitive-equation numerical model (section 2). An exhaustive study is not attempted. Instead, a few calculations are used to show how ambient alongshelf currents can change the dense water flux and properties over a uniform shelf (section 3.1) and to test the possibility that submarine canyons may provide preferred pathways of dense water transport (section 3.2). The results are then discussed and summarized (section 4).

2. Numerical Model

The s-coordinate primitive-equation model (SPEM-5.1), developed by D. Haidvogel's group at Rutgers University, is used to solve the standard hydrostatic and Boussinesq momentum, density, and continuity equations:

$$u_t + \mathbf{u}\cdot\nabla u - fv = -\frac{1}{\rho_0}p_x + (A_V u_z)_z + F_u \quad (1)$$

$$v_t + \mathbf{u}\cdot\nabla v + fu = -\frac{1}{\rho_0}p_y + (A_V v_z)_z + F_v \quad (2)$$

$$p_z = -\rho g \quad (3)$$

$$\rho_t + \mathbf{u}\cdot\nabla\rho = (K_V\rho_z)_z + F_\rho \quad (4)$$

$$\nabla\cdot\mathbf{u} = 0. \quad (5)$$

In this system (u, v, w) represent the (x, y, z) components of the velocity $\mathbf{u}$ with z directed vertically upward from the ocean surface, ρ is the difference between the total density and a constant reference density ρ_0, p is pressure, f is the Coriolis parameter, g is gravitational acceleration, A_V is the vertical eddy viscosity, K_V is the vertical eddy diffusivity, and t is time. Variables $F_{u,v,\rho}$ represent dissipative functions that are required for numerical stability. Subscripts x, y, z, and t denote partial differentiation. SPEM5.1 is similar to earlier versions of SPEM [e.g., *Haidvogel et al.*, 1991], the most important differences being that SPEM5.1 uses finite differences in both horizontal and vertical directions and a generalized stretched vertical coordinate (the s coordinate described by *Song and Haidvogel* [1994]) to allow greater flexibility in specifying the locations of vertical grid points. For the present calculations, the grid points are uniformly distributed between surface and bottom.

At the surface, the momentum flux is zero (i.e., no stress), while a density (i.e. buoyancy) flux is prescribed to represent brine rejection accompanying ice formation in the coastal polynya (see below). At the bottom, the density flux is zero, while the shear stress is specified by a linear bottom stress parameterization:

$$A_V u_z = ru; \quad A_V v_z = rv \quad \text{at} \quad z = -h, \quad (6)$$

where $h(x, y)$ is the bottom depth and r is a bottom friction coefficient, taken here to be 10^{-4} m s^{-1}. There is no flux of density or momentum at solid model sidewalls.

The vertical mixing coefficients A_V and K_V are approximated by using the Mellor-Yamada level 2 turbulence closure scheme as applied in SPEM5.1. This is a slightly simplified version of the standard scheme in that the maximum mixing length scale is assumed independent of z to avoid the need for iterations. Also, a maximum value of 10^{-3} m^2 s^{-1} is imposed to reduce numerical errors in the finite difference approximations of the vertical mixing terms. The minimum value for A_V and K_V is 10^{-5} m^2 s^{-1}. Additional vertical mixing of density is applied in the form of instantaneous convective adjustment whenever the water column becomes statically unstable (i.e., when lighter water appears under heavier water). For numerical stability, Laplacian subgrid-scale mixing with constant mixing coefficients is applied along horizontal surfaces (i.e., $F_{u,v,\rho} = \nu_{u,v,\rho}\nabla^2 u, v, \rho$), using the smallest mixing coefficients that produce numerically stable calculations, $\nu_{u,v} = 20$ m^2 s^{-1} and $\nu_\rho = 5$ m^2 s^{-1}.

The model domain is a high-latitude uniformly rotating ($f = 1.3 \times 10^{-4}$ s^{-1}) rectangular channel with solid walls at the coast ($y = 0$) and offshore ($y = 75$ km) and open cross-shelf boundaries at $x = 0$ and $x = 300$ km. Grid spacing is about 1 km in each horizontal direction (288 by 72 grid points), with 20 points in the vertical. At the upstream boundary ($x = 0$), a uniform and constant alongshelf inflow with magnitude U is prescribed (Figure 1). At the downstream boundary ($x = 300$ km), a simple advection condition is applied as follows. At each time step, SPEM5.1 first solves for the barotropic vorticity, which is then used to find the barotropic stream function, ψ. Subsequently, the depth-varying velocities and the density are computed. The advection condition consists, at each time step, of giving the barotropic vorticity at the downstream boundary the

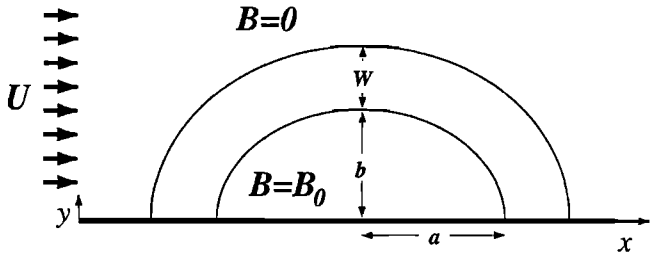

Figure 1. Idealized polynya geometry and surface forcing. Constant surface buoyancy flux B_0 is imposed within the half-ellipse with length $2a$ and offshore width b along the coast ($y = 0$). Buoyancy flux decreases linearly to zero across the forcing decay region with width W. An ambient alongshelf current U is imposed at $x = 0$. For all calculations presented here, $B_0 = 4 \times 10^{-7}$ m^2 s^{-3}, $a = 25$ km, $b = 10$ km, and $W = 10$ km.

value from the closest interior grid point (i.e., simple advection in x). Next, the barotropic stream function at the boundary is computed from the vorticity equation after assuming that $\psi_{xx} = 0$. (This requires a straightforward tridiagonal matrix inversion.) Then the density and depth-varying velocities at the boundary are replaced by the values from the closest interior grid points. This condition produces better results than a wave-like radiation condition and/or a sponge because of the advective nature of the interior flows. The bottom depth increases uniformly offshore from a minimum of 50 m at the coast, with bottom slope 10^{-3}. Some calculations include a submarine canyon that transects the topography either normally, diagonally, or parallel to the coast (see section 3.2 for details).

The calculations closely follow those of *Chapman and Gawarkiewicz* [1997], except for the imposed ambient alongshelf current. Each calculation begins from rest with a homogeneous ocean with density ρ_0. At time $t = 0$, the alongshelf inflow begins and rapidly adjusts to the topography. Coincidentally, the surface buoyancy flux is applied (Figure 1): a constant value of B_0 within the half-ellipse of alongshelf extent $2a$ and offshore width b, surrounded by a forcing decay region of width W over which the surface buoyancy flux decreases linearly from B_0 to zero. Both the inflow and the surface forcing remain imposed throughout the duration of each calculation, typically 25 days of simulation time. The time step is 432 s.

3. Results

3.1. Uniform Topography

The focus here is on the influence of the ambient alongshelf current imposed at $x = 0$, so the bottom topography and the parameters defining the surface buoyancy flux remain the same throughout, while the inflow velocity U is changed. The bottom topography is given by

$$h = h_c + 0.001y, \tag{7}$$

where $h_c = 50$ m is the depth at the coast. The polynya dimensions (Figure 1) are $a = 25$ km, $b = 10$ km, and $W = 10$ km, while the maximum buoyancy flux is $B_0 = 4 \times 10^{-7}$ m^2 s^{-3}. These choices are arbitrary but not unreasonable for Arctic polynyas. The slightly large buoyancy flux reduces the theoretical time to reach equilibrium, thus shortening the simulation time required for each calculation.

The ocean response with no ambient current ($U = 0$) is shown in Figure 2. The sequence of snapshots of the bottom density anomaly is very much like those of previous studies [e.g., *Gawarkiewicz and Chapman*, 1995; *Chapman and Gawarkiewicz*, 1995, 1997; *Chapman*, 1999]. Eddies have formed by day 10 and carry dense water offshore, with little alongshelf flux. By day 20, dense water has nearly reached the offshore boundary, but only a small plume of dense water, trapped to the coast, has moved alongshelf past $x = 150$ km.

The response with an ambient alongshelf flow of $U = 0.1$ m s^{-1} appears in Figure 3 for comparison. The ambient current sweeps much of the dense water downstream of the polynya within 5 days. However, the instabilities are qualitatively unchanged; i.e., they develop in about the same time and have roughly the same scales. The most noticeable difference is that the eddies are carried downstream by the ambient current; so the dense water flux is alongshelf as well as offshore. By day 20, dense water has traveled beyond the end of the model domain, while still nearly reaching the offshore boundary.

The alongshelf density flux ($\int_0^{75km} u\rho\, dy$) measured at $x = 150$ km is plotted in Figure 4 for the two calculations described above and another with $U = 0.2$ m s^{-1}. As expected, the alongshelf density flux is nearly absent without an ambient current. With an ambient current, the density fluxes are comparable, but the dense water arrives sooner for larger U. This makes sense because the total change in density (i.e., buoyancy input) depends only on the surface buoyancy flux, which is the same for both calculations, and not on the ambient current. A swifter current reduces the density anomalies because the time of exposure to the surface forcing is reduced, but it exposes more water to the forcing because of the larger velocities. These effects (decreased ρ, increased u) tend to compensate to produce similar density anomaly fluxes.

Figure 5 shows the time dependence of the average density anomaly near the coast within the polynya for different ambient currents. With no ambient current ($U = 0$), the density increases linearly until eddy fluxes carry dense water out of the polynya (about day 6), after which the density anomaly is quasi-steady, with variations caused by eddies moving into and out of the polynya (as in previous studies). A weak alongshelf flow ($U = 0.05$ m s^{-1}) reduces both the maximum density anomaly and the subsequent variations because some

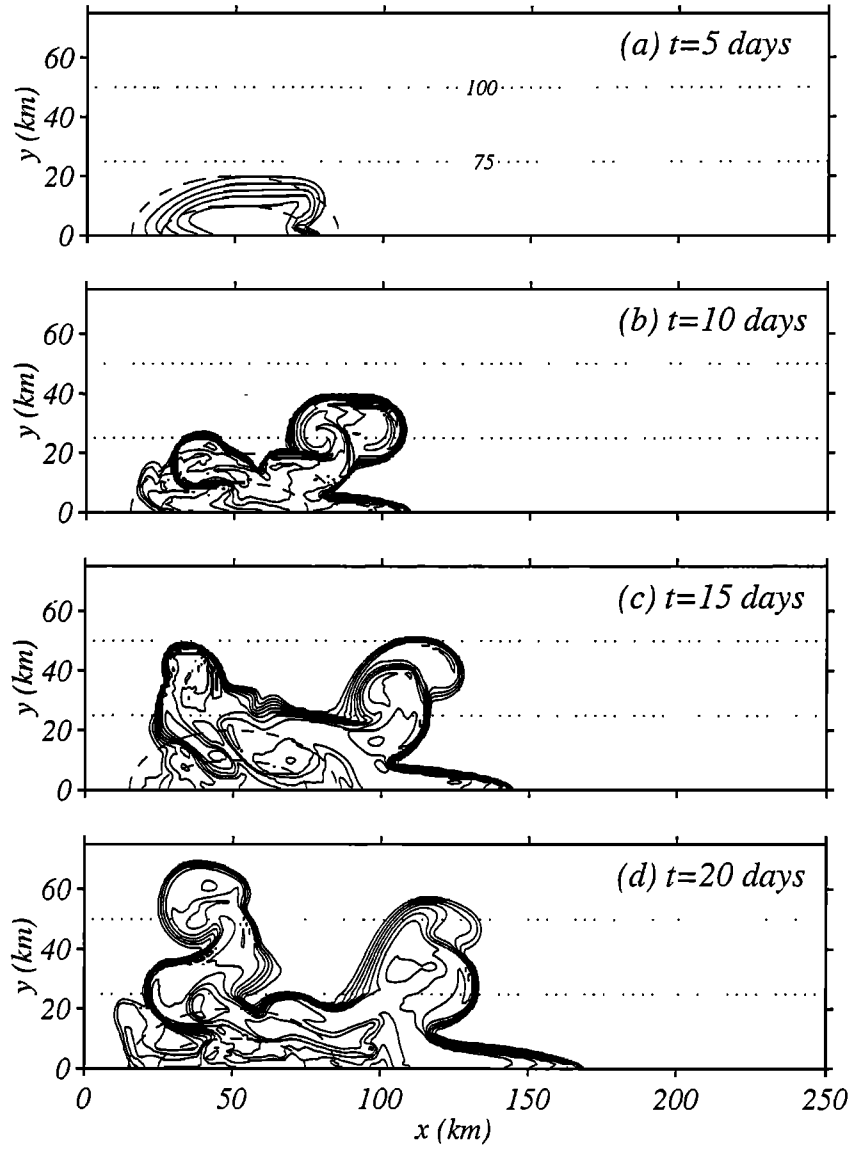

Figure 2. Plan views of density anomaly at the bottom after 5, 10, 15, and 20 days with no ambient current ($U = 0$). Contours are 0.06 to 0.6 by 0.06 kg m^{-3}, with the smallest values farthest from the coast. Dashed curves denote the polynya forcing region as in Figure 1. Dotted lines are the 75- and 100-m isobaths.

of the eddy variability is swept downstream out of the polynya. Stronger ambient currents ($U = 0.1$, 0.2 m s^{-1}) further reduce the maximum density anomaly and eliminate the variability at longer times. In these cases, the ambient current is rapid enough to carry water out of the polynya before the onset of eddy exchange. That is, the advection time to pass through the polynya, $2a/U$, is less than the time needed for eddy exchange to fully develop (about 6 days for the present cases). The advection time for $U = 0.1$ m s^{-1} is 5.8 days; so no eddies should be found within the forcing region. Indeed, Figure 3 shows that the dense water eddies appear only downstream of the polynya.

In general, all eddy activity occurs downstream of the polynya when the advection time is less than the time to reach equilibrium (t_e given by *Chapman and Gawarkiewicz* [1997]);

$$2a/U < t_e = \beta(fWb/B_0)^{1/2}$$

or

$$U > (2a/\beta)(B_0/fWb)^{1/2}, \qquad (8)$$

where β is an empirical constant (roughly 3 based on the $U = 0$ curve in Figure 5). A longer (larger a) or narrower (smaller b) polynya with stronger forcing (larger B_0) requires a faster current to satisfy (8); so ambient alongshelf currents should have greater impact on shorter, wider polynyas with weak surface buoyancy fluxes. For the chosen parameters, (8) is satisfied when $U > 0.092$ m s^{-1}, consistent with Figure 5.

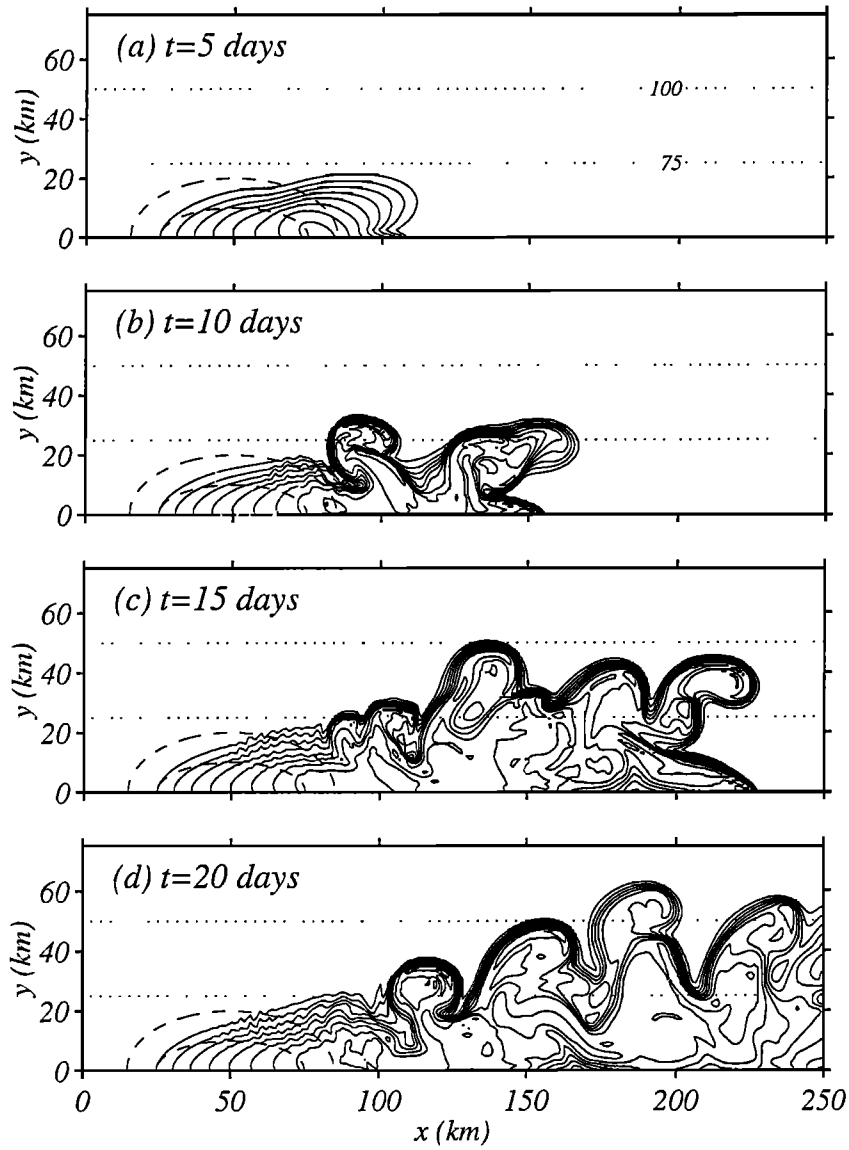

Figure 3. As in Figure 2, but with an ambient current of $U = 0.1$ m s^{-1}. Contours are 0.04 to 0.4 by 0.04 kg m^{-3}.

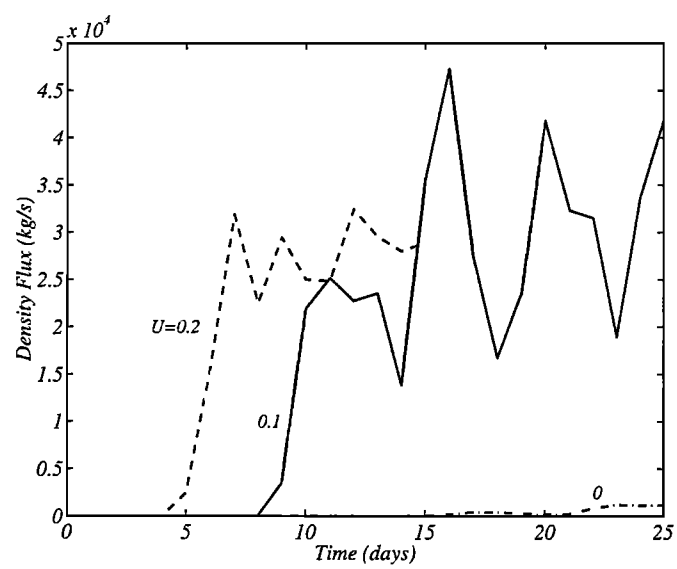

Figure 4. Alongshelf density flux ($\int_0^{75km} u\rho\, dy$) at $x = 150$ km for the standard calculation with uniform topography and $U = 0$, 0.1, and 0.2 m s^{-1}.

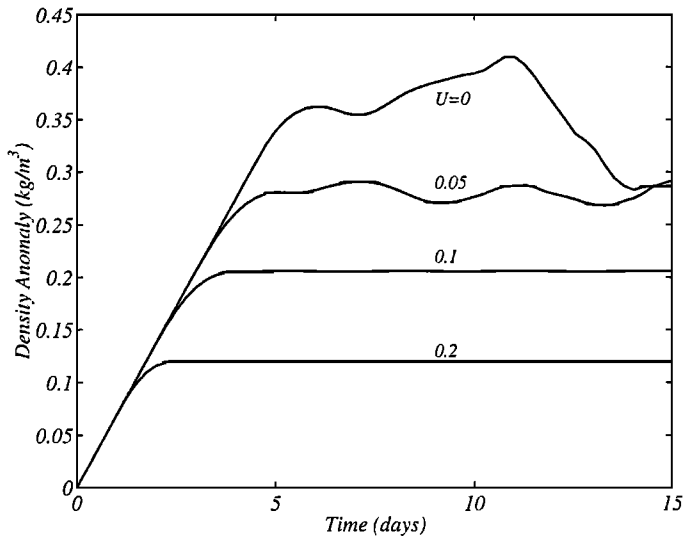

Figure 5. Average bottom density anomaly near the coast within the polynya for the standard calculation with uniform topography and $U = 0$, 0.05, 0.1, and 0.2 m s^{-1}.

The ambient alongshelf current also changes the volume distribution of density anomalies. Figure 6 shows time histories of the volume of densified water within various density anomaly ranges (indicated by the bar on the right). Only density anomalies greater than 0.1 kg m^{-3} are included. In all cases, the total volume of densified water continues to increase with time, although individual density classes may increase or decrease. With no ambient current ($U = 0$), a wide range of density anomalies forms, the maximum appearing at about day 10 in small quantities between 0.575 and 0.8 kg m^{-3} and disappearing by day 20. Interestingly, most of the dense water (about 2/3 in this case) occurs between 0.1 and 0.325 kg m^{-3}, appreciably less than the equilibrium density anomaly of about 0.35 kg m^{-3} (Figure 5). This suggests caution when using the simple formulas from previous studies [e.g., *Chapman and Gawarkiewicz*, 1997] to estimate volume fluxes of dense water. The majority of the dense water may actually have smaller density anomalies than the predicted equilibrium density anomaly. This may help to explain the substantial reduction in density anomalies found in dense water eddies that reach the shelf break [*Gawarkiewicz*, 2000]; i.e., these eddies may have formed within water of lesser density anomalies.

An ambient alongshelf current reduces the maximum density anomalies but increases the volume of densified waters (as argued above). For $U = 0.05$ m s^{-1}, no water in the maximum density anomaly class is formed; the largest density anomalies are between 0.45 and 0.575 kg m^{-3}, and this range disappears by day 20. The vast majority of the dense water is found between 0.1 and 0.325 kg m^{-3}. For $U = 0.1$ m s^{-1}, no water with density anomaly greater than 0.45 kg m^{-3} is formed. For $U = 0.2$ m s^{-1} (not shown), the maximum is reduced to below 0.325 kg m^{-3}. The point here is that a wide range of density anomalies is produced by a coastal polynya, and the range is sensitive to the strength of the ambient currents.

If there were no eddies and no horizontal mixing, then the water would simply be advected beneath the polynya with the density anomaly at any horizontal location and time given by (assuming complete vertical mixing)

$$\rho(x, y, t) = \frac{\rho_0}{gHU} \int_{x-Ut}^{x} B(x', y)\, dx', \qquad (9)$$

where B is the surface buoyancy flux shown in Figure 1. In this case, density anomalies and water volumes in various density anomaly ranges can be calculated directly from (9). An example of the density anomaly field that would be formed (with the standard forcing) after 10 days with $U = 0.1$ m s^{-1} is shown in Figure 7. The effects of eddies can be seen by comparing Figure 7 with Figure 3b. Figure 8 compares the volume distribution of density anomalies from the numerical calculations (Figure 6) with those based on advection alone. Each set of curves (solid curves for the numerical model and dashed curves for pure advection) is numbered to correspond with the density anomaly ranges used in Figure 6, with 1 being the lowest range. Deviations between solid and dashed curves in each set represent eddy and mixing effects. Note that no water in range 4 (0.45 to 0.575 kg m^{-3}) is formed in the numerical model with $U = 0.1$ m s^{-1}. For both ambient currents, the numerical model distributions initially tend to follow the advection curves. However, after about 10 days, the curves diverge with the numerical model, producing more water at smaller density anomalies and less water at larger density anomalies. Clearly, these changes are brought about by eddies and mixing, although the details are difficult to sort out quantitatively.

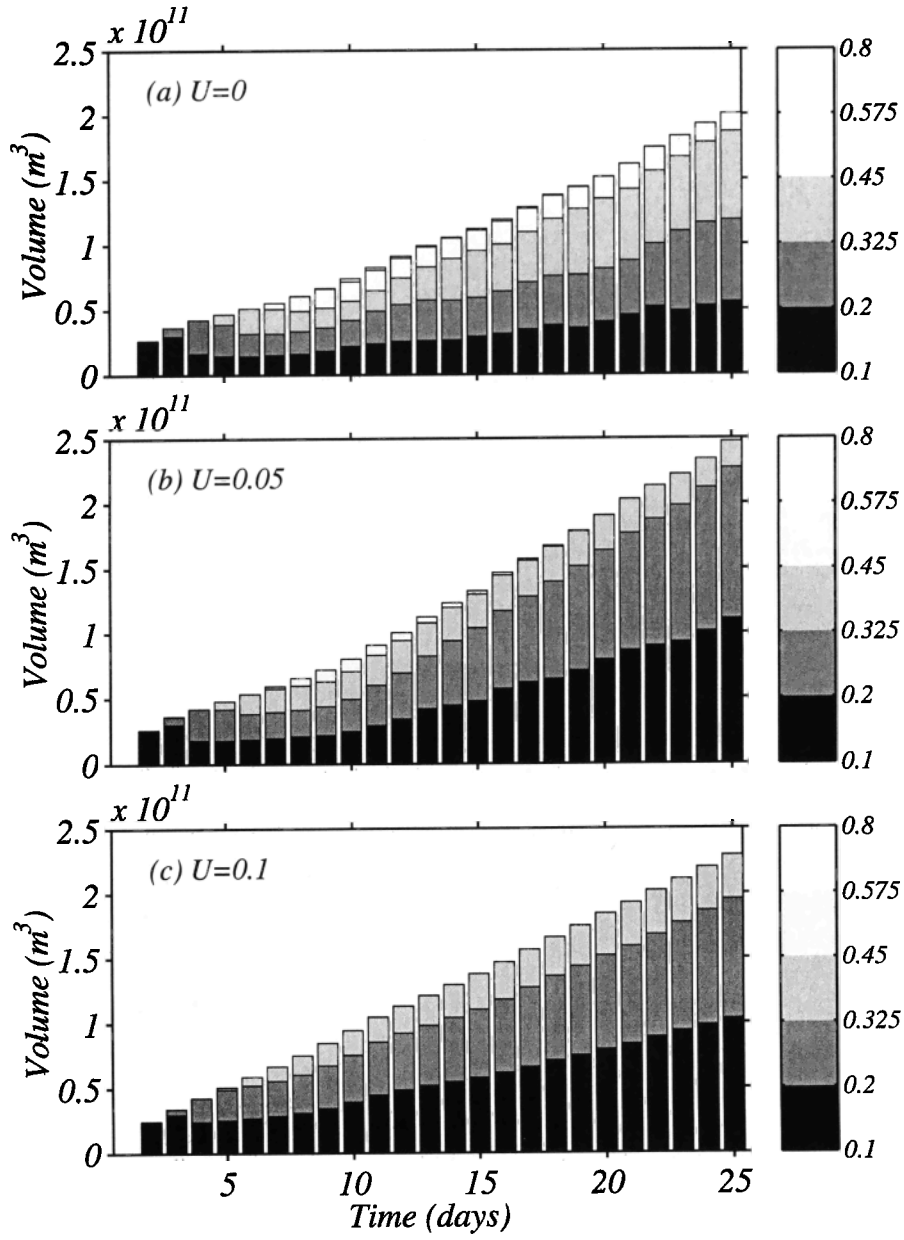

Figure 6. Time histories of the volume of densified water within various ranges of density anomalies (indicated in kilograms per cubic meter by the bar on the right) for the standard calculation with uniform topography and (a) $U = 0$, (b) $U = 0.05$ m s^{-1}, and (c) $U = 0.1$ m s^{-1}.

3.2. Submarine Canyons

Continental shelves are not generally uniform; so an ambient alongshelf current could carry dense water into the proximity of topographic variations, e.g., a submarine canyon, where the dense water might descend rapidly and leave the shelf. To investigate this possibility, several of the above calculations have been repeated with topography that includes a Gaussian-shaped submarine canyon. Three canyon orientations are considered, and in each case the canyon is located far enough downstream that little dense water would reach the canyon if there were no ambient alongshelf current (e.g., $x > 125$ km in Figure 2).

The first two canyons considered are oriented normally and diagonally to the coast. Their respective topographies are given by

$$h = h_c + 0.001y[1 + e^{-(x-125\times10^3\mathrm{m})^2/25\times10^6\mathrm{m}^2}] \quad (10)$$

and

$$h = h_c + 0.001y[1 + e^{-(x-y-125\times10^3\mathrm{m})^2/25\times10^6\mathrm{m}^2}]. \quad (11)$$

In each case, the coastal depth remains at $h_c = 50$ m, while the maximum depth is set at 200 m.

Figure 9 shows the density anomaly at the bottom for each topography after 15 days with an ambient along-

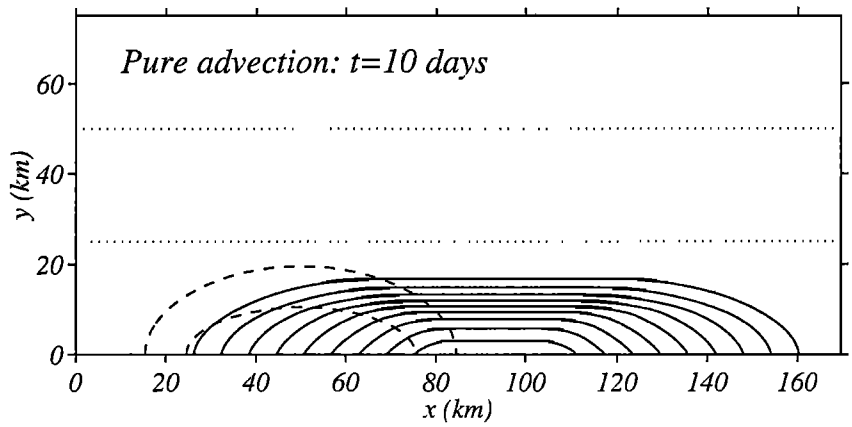

Figure 7. Plan view of density anomaly at the bottom after 10 days of pure advection (i.e., no eddies or mixing) using the standard parameters with an ambient current of $U = 0.1$ m s^{-1}. Contours are 0.04 to 0.4 by 0.04 kg m^{-3}. Compare with Figure 3b.

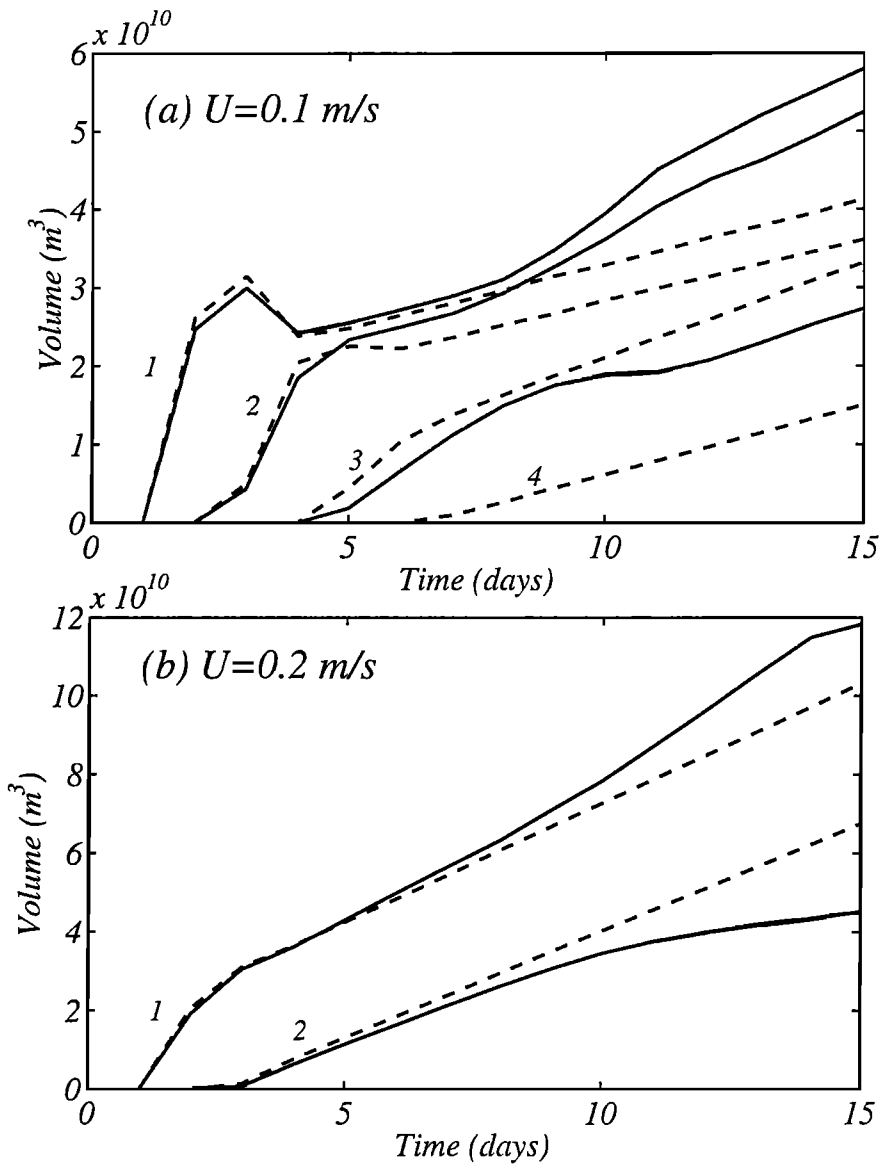

Figure 8. Time histories of the volume of densified water within the density anomaly ranges in Figure 6 (with 1 being the lowest range) for the numerical calculations (solid curves) and pure advection (dashed curves). Standard parameters are used with (a) $U = 0.1$ m s^{-1} and (b) $U = 0.2$ m s^{-1}.

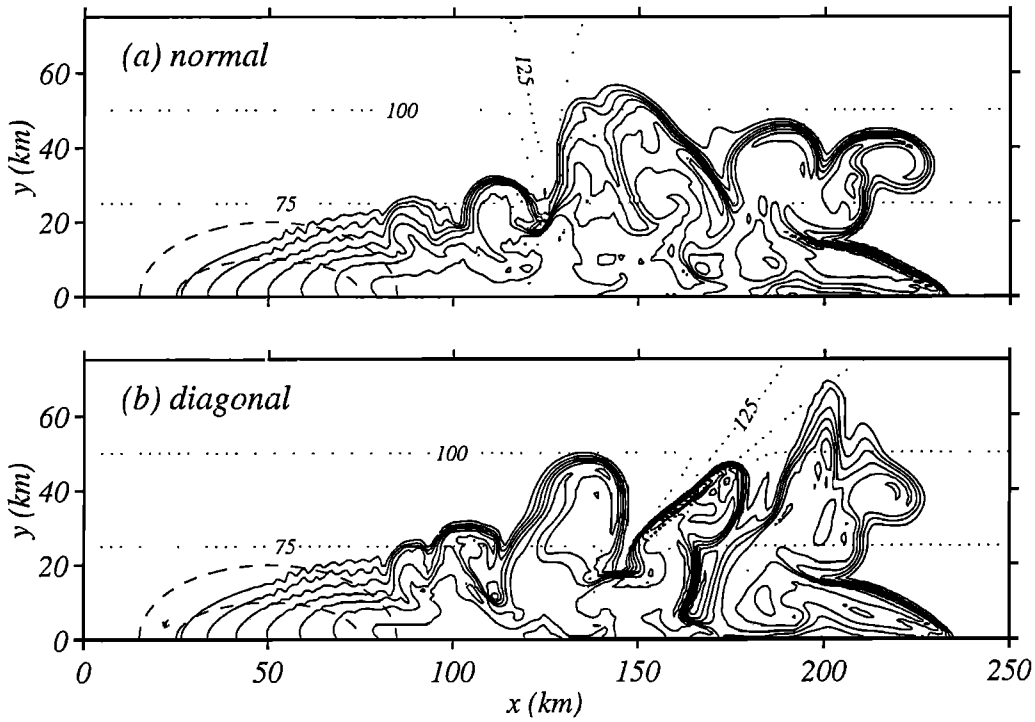

Figure 9. As in Figure 3c (after 15 days), but with a submarine canyon transecting the shelf downstream of the polynya. The 75-, 100-, and 125-m isobaths are shown. Contours are 0.05 to 0.5 by 0.05 kg m^{-3}.

shelf current of $U = 0.1$ m s^{-1}. The most obvious feature to note is that almost no dense water is found in either canyon, despite large quantities of dense water appearing farther downstream ($x > 125$ km). In fact, the dense water almost appears to avoid the canyon. The reason is simple: the ambient alongshelf flow is nearly geostrophic with a strong tendency to follow isobaths; only bottom friction produces significant cross-isobath currents. Therefore the flow moves around the canyon, accelerating as it squeezes between the coast and the canyon head (Figure 10), carrying the dense water with it and thereby inhibiting the dense water from flowing into the canyon. Virtually all of the dense water is eventually carried past the canyon; so the alongshelf density flux (not shown) is comparable to that without a canyon (Figure 4), although the details are different because of differences in eddy behavior.

The third canyon orientation is parallel to the coast, which may be considered a crude representation of Barrow Canyon on the northeast Chukchi Shelf. The topography is given by

$$h = h_c + 0.001[y + H(x - 125 \times 10^3\text{m}) \\ (x - 125 \times 10^3\text{m})e^{-(y-20\times10^3 m)^2/25\times10^6\text{m}^2}], \quad (12)$$

where $H(x)$ is the Heaviside step function, and the canyon axis is located 20 km offshore. Although the idealized canyon in (12) is not identical to Barrow Canyon, the axis bottom slope and the maximum slope of the canyon walls (~0.01) are comparable; so the topographic influence should be similar. In this case, the ambient alongshelf current carries dense water parallel to the canyon axis, and a substantial quantity of dense water flows into the canyon (Figure 11). The tendency for the flow to follow isobaths still causes the current to accelerate between the coast and the canyon, creating large velocities near the coast and a region of greatly reduced (and even reversed; $u < 0$) velocities over the offshore side of the canyon (Figure 12). This velocity structure bears some resemblence to flows in Barrow Canyon recently observed [e.g., *Münchow and Carmack*, 1997, Figure 12] and modeled [*Signorini et al.*, 1997].

The flux of dense water that moves down the canyon relative to that which is carried near the coast past the canyon depends strongly on the strength of the ambient current. Figure 13 shows that roughly half the density flux is in the canyon when $U = 0.1$ m s^{-1}, while very little density flux occurs in the canyon with a faster ambient current of $U = 0.2$ m s^{-1} because the currents shoreward of the canyon are much stronger. (Note that the transport of ambient water does not contribute to the density flux calculation because the ambient water has zero density anomaly, i.e., $\rho = 0$.) Not surprisingly, point measurements of density anomaly and alongshelf velocity within the canyon and on the adjacent shelf (Figure 14) are highly variable as eddies pass, with currents ranging from 0.3 to 0.5 m s^{-1}. The details are, of course, sensitive to the canyon location; i.e., if the canyon is closer to the coast, then more dense water flows into the canyon.

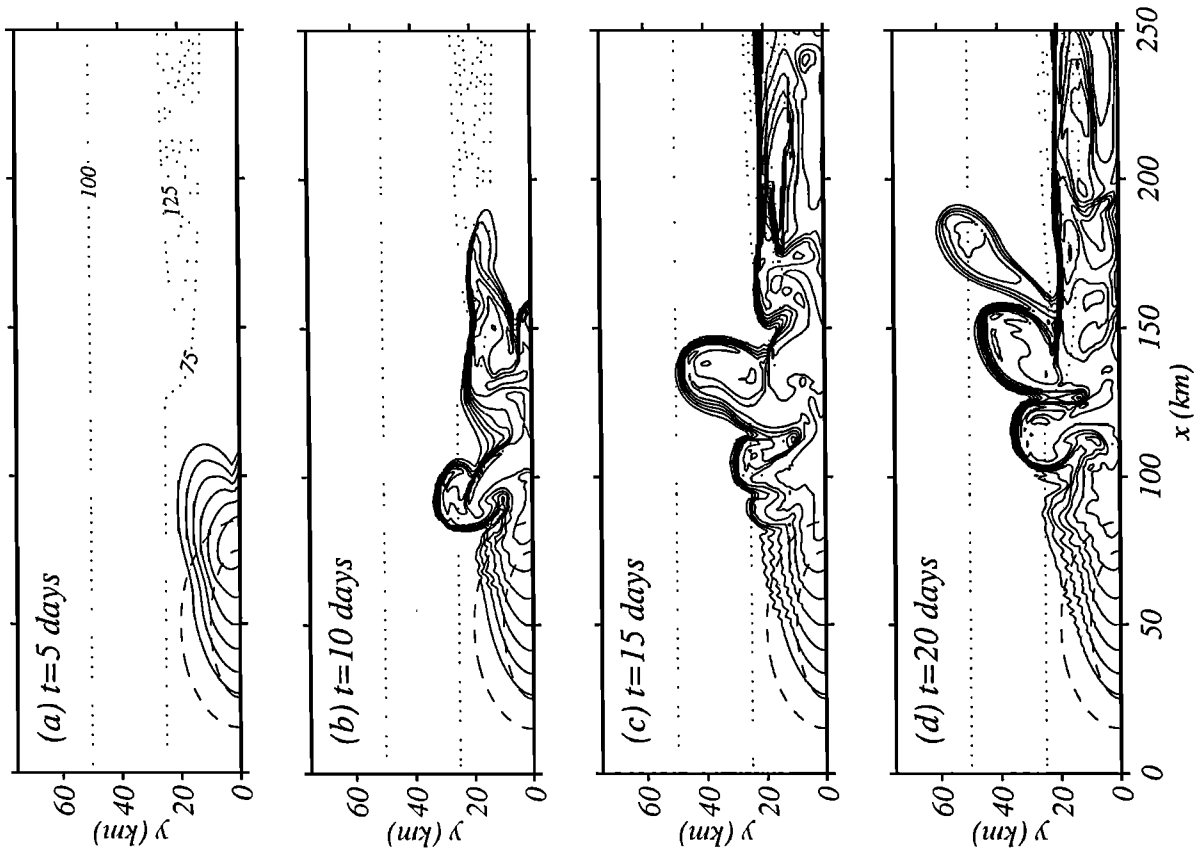

Figure 11. As in Figure 3, but with a canyon oriented parallel to the coast. Contours are 0.05 to 0.5 by 0.05 kg m^{-3}

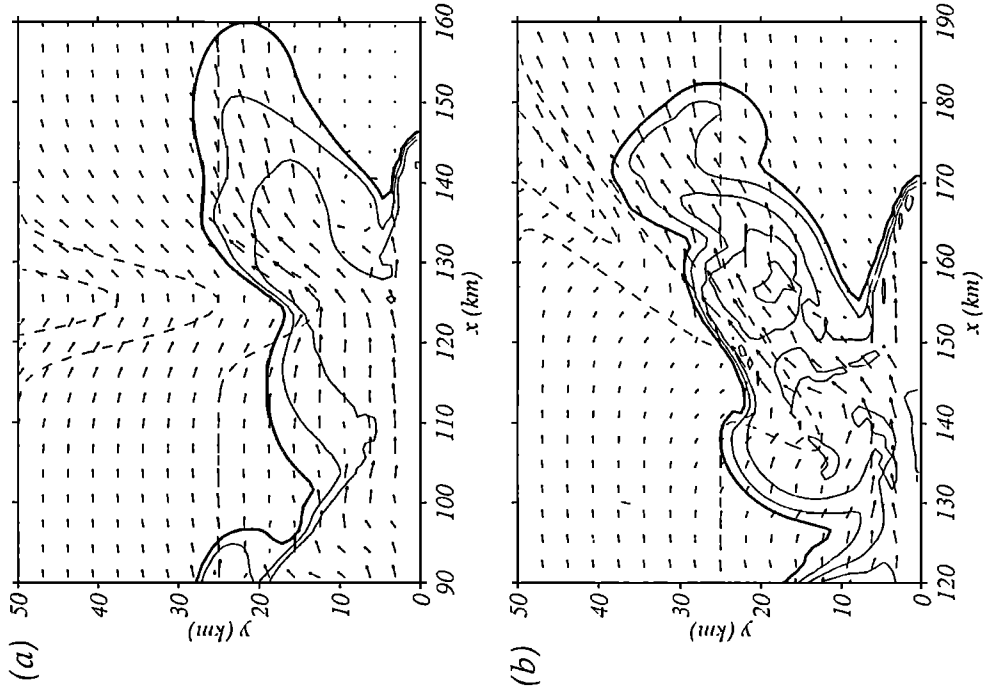

Figure 10. Plan views of density anomaly and horizontal velocity vectors at the bottom with (a) normally and (b) diagonally oriented canyons (see Figure 9). Vectors are plotted every third grid point. Only a small region of the model domain near the canyon head is shown. Contours are 0.1 to 0.4 by 0.1 kg m^{-3}, with 0.1 kg m^{-3} denoted by the thicker contour. In Figure 10a, $t = 9.25$ days and maximum velocity is 0.275 m s^{-1}. In Figure 10b, $t = 11$ days and maximum velocity is 0.32 m s^{-1}. Dashed curves are the 75-, 100-, and 125-m isobaths.

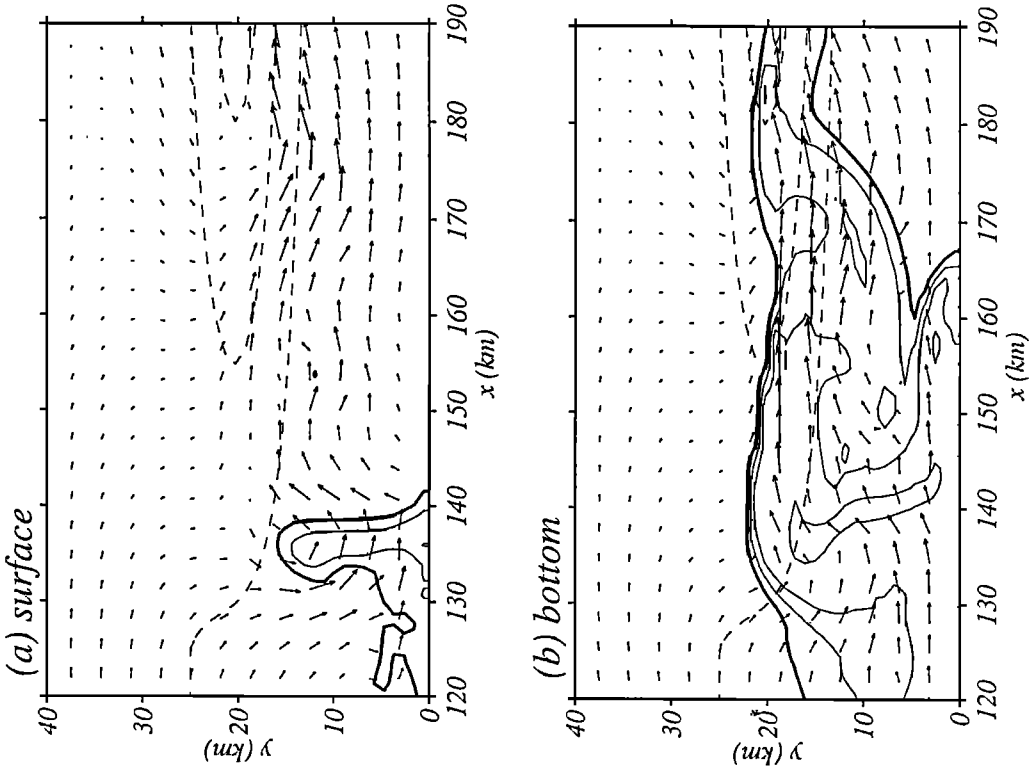

Figure 12. Plan views of density anomaly and horizontal velocity vectors at the (a) surface and (b) bottom after 10.5 days with the parallel canyon (see Figure 11). Vectors are plotted every third grid point. Only a small region of the model domain near the canyon head is shown. Contours are 0.1 to 0.4 by 0.1 kg m^{-3} with 0.1 kg m^{-3} denoted by the thicker contour. Maximum velocity is 0.43 m s^{-1}. Dashed curves are the 75-, 100-, and 125-m isobaths.

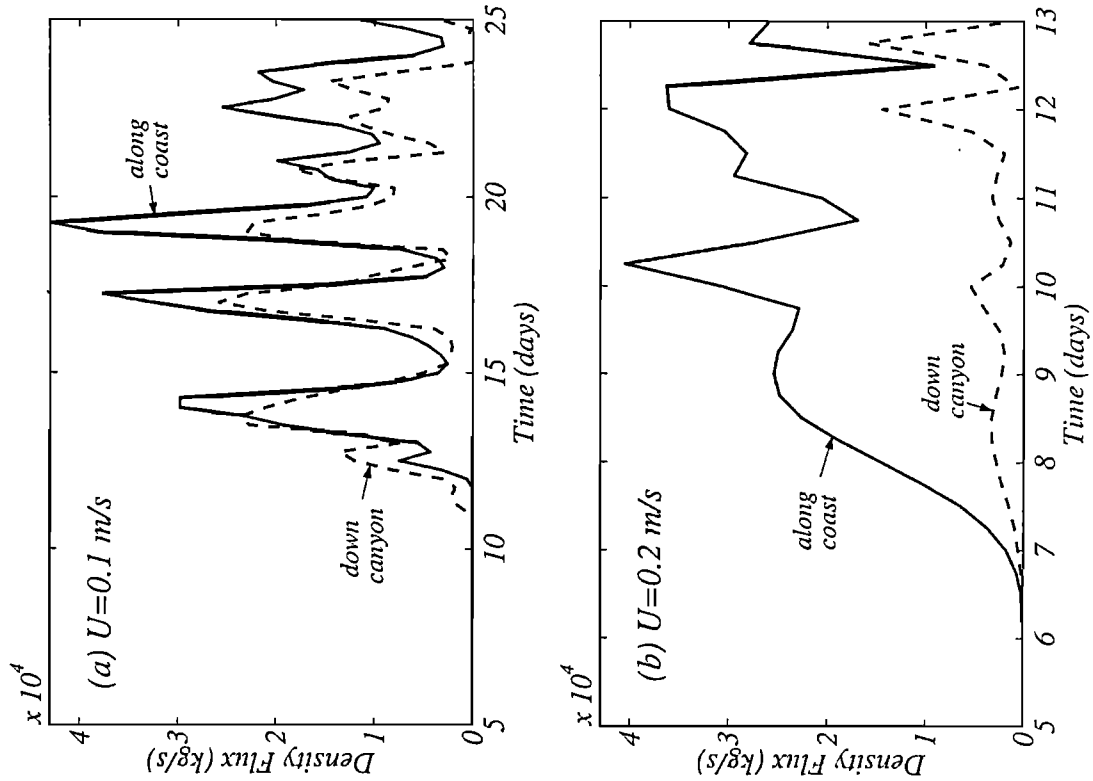

Figure 13. Alongshelf density flux for calculations with the parallel canyon, measured at $x = 220$ km (see Figure 11). Solid curves are density fluxes between the coast and the canyon ($0 < y < 13$ km); dashed curves are fluxes down the canyon (13 km $< y < 28$ km). Standard parameters are used with (a) $U = 0.1$ m s^{-1} and (b) $U = 0.2$ m s^{-1}.

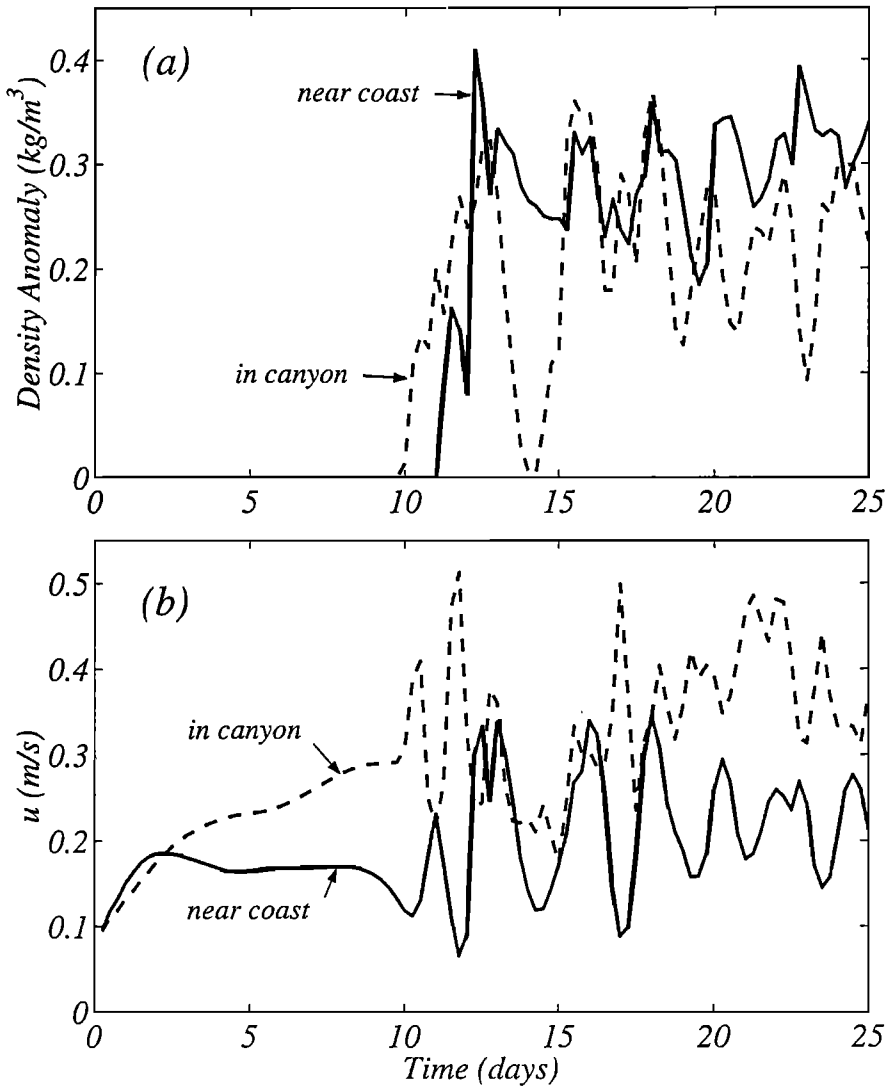

Figure 14. Time series of (a) density anomaly and (b) alongshelf velocity for the calculation shown in Figures 11 and 12, measured at $x = 190$ km and (solid curves) near the coast at $y = 5$ km, $z = 1.3$ m above the bottom, and (dashed curves) in the canyon at $y = 17$ km, $z = 2$ m above the bottom.

4. Discussion and Summary

The small number of results presented here by no means constitutes a thorough study of the effects of ambient currents on the formation and offshore transport of dense water from a coastal polynya. The parameter space is enormous if variations in current structure, shelf topography, polynya geometry, surface forcing, etc., are considered. The details of the present results will change as these parameters are changed. The intent here is merely to demonstrate the importance of ambient alongshelf currents and to suggest ways that ambient currents may alter dense water properties and transport pathways on high-latitude shelves.

Ambient alongshelf currents do not change the fundamental processes by which dense water moves offshore from the polynya; i.e., instabilities generate eddies that carry dense water away. However, ambient currents change both the density anomalies generated and the direction of eddy transport. Without ambient currents, density flux is primarily offshore, with little flux along the shelf. An ambient alongshelf current carries eddies downstream and reduces the time that water parcels are exposed to surface forcing in the polynya. Thus the maximum density anomalies are reduced, although the volume of densified water is somewhat larger because more water is exposed to surface forcing. These effects will be less important for a longer polynya because water parcels are more likely to remain beneath the polynya until eddy processes dominate. It is important to note that a wide range of density anomalies is formed regardless of the ambient current. This result is qualitatively consistent with the wide range of water properties reported by *Weingartner et al.* [1998, Figure 7] and suggests caution when estimating a "typical" density anomaly generated from a coastal polynya.

Ambient alongshelf currents are also important in determining the transport pathways of dense water, but perhaps not entirely as expected. It has sometimes been assumed that dense water will preferentially descend down submarine canyons and that ambient currents might enhance such flows by carrying dense water to the canyons. However, the tendency for ambient shelf flows to follow isobaths acts against the gravitational acceleration down the canyon; so swift ambient currents can actually reduce the flow of dense water down a canyon. Dense water descent down submarine canyons appears to be most likely in the absence of strong ambient currents when dense water is formed in the vicinity of a canyon, e.g., Storfjord Trough [*Quadfasel et al.*, 1988; *Honjo et al.*, 1988], or in cases where the canyon is oriented parallel to the ambient current, e.g., Barrow Canyon. However, even in the case of Barrow Canyon, *Münchow and Carmack* [1997] present evidence of substantial flow past Barrow Canyon onto the Beaufort Shelf, suggesting the possibility that dense water could bypass the canyon.

Taken together, these results suggest that ambient currents could provide an important mechanism for variability of dense water formation and offshore transport both within a winter and interannually. When ambient currents are weak or absent, density anomalies should be largest and the flux should be primarily offshore. Periods with strong ambient currents will tend to produce smaller density anomalies, and the density flux should be in the direction of the ambient currents. In the case of the Chukchi Shelf, *Weingartner et al.* [1998] have already reported this connection between the presence of alongshelf currents and dense water flow down Barrow Canyon. Similarly, ambient currents could account for the interannual variability of dense water flux in Barrow Canyon; i.e., in some years it is found, while in others it is not found [e.g., *Garrison and Becker*, 1976; *Aagaard et al.*, 1985; *Aagaard and Roach*, 1990].

Although not explicitly addressed here, the results suggest that the ultimate fate of eddies that move offshore and reach the shelf break may be strongly dependent on the presence or absence of a boundary current located near the shelf break. Such a current could inhibit eddies from entering the deep basin, or at least alter their location of entry. These effects remain to be studied.

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D. C. Chapman, Clark 316B, MS 21, Woods Hole Oceanographic Institution, Woods Hole, MA 02543. (dchapman@whoi.edu)

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